

The Tangent Approximation

1. The Tangent Plane

For a function of one variable, $w = f(x)$, the tangent line to its graph at a point (x_0, w_0) is the line passing through (x_0, w_0) and having slope

$$\left. \frac{dw}{dx} \right|_{x=x_0}.$$

For a function of two variables,

$$w = f(x, y),$$

the natural analogue is the tangent plane to the graph at the point

$$(x_0, y_0, w_0),$$

where

$$w_0 = f(x_0, y_0).$$

The tangent plane:

- must pass through the point (x_0, y_0, w_0) ;
- must contain the tangent lines to the surface in the x - and y -directions.

Hence, the equation of the tangent plane is

$$w - w_0 = \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0),$$

or equivalently,

$$w = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

The tangent plane contains the tangent lines to the graphs of the two partial functions. This will hold if the plane has the same slopes in the i - and j -directions as the surface does.

Using these two conditions, it is easy to find the equation of the tangent plane. The general equation of a plane through (x_0, y_0, w_0) is

$$A(x - x_0) + B(y - y_0) + C(w - w_0) = 0.$$

Assume the plane passes through (x_0, y_0, w_0) . What values of the coefficients A and B will make it also tangent to the graph?

Let

$$a = \text{slope of the plane in the } i\text{-direction.}$$

By putting

$$y = y_0$$

in the equation of the plane, the slope of the plane in the i -direction is

$$a = \left(\frac{\partial w}{\partial x} \right)_0,$$

which is the slope of the graph in the i -direction by the definition of partial derivatives.

Similarly,

$$b = \left(\frac{\partial w}{\partial y} \right)_0.$$

Hence, the tangent plane can be written as

$$w - w_0 = \left(\frac{\partial w}{\partial x} \right)_0 (x - x_0) + \left(\frac{\partial w}{\partial y} \right)_0 (y - y_0).$$

Therefore, the equation of the tangent plane to

$$w = f(x, y)$$

at the point (x_0, y_0) is

$$w - w_0 = \left(\frac{\partial w}{\partial x} \right)_0 (x - x_0) + \left(\frac{\partial w}{\partial y} \right)_0 (y - y_0),$$

where

$$w_0 = f(x_0, y_0).$$

2. The Approximation Formula

The most important use of the tangent plane is to give an approximation that is the basic formula in the study of functions of several variables. Almost everything follows in one way or another from it.

Tangent Approximation

The intuitive idea is that if we stay near the point

$$(x_0, y_0, w_0),$$

the graph of the tangent plane will be a good approximation to the graph of the function

$$w = f(x, y).$$

Therefore, if the point (x, y) is close to (x_0, y_0) ,

$$f(x, y) \approx w_0 + \left(\frac{\partial w}{\partial x}\right)_0 (x - x_0) + \left(\frac{\partial w}{\partial y}\right)_0 (y - y_0).$$

Equivalently,

$$f(x, y) \approx f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Tangent Plane and Tangent Approximation

The equation of the tangent plane to $w = f(x, y)$ at (x_0, y_0, w_0) is

$$w - w_0 = \left(\frac{\partial w}{\partial x}\right)_0 (x - x_0) + \left(\frac{\partial w}{\partial y}\right)_0 (y - y_0).$$

The most important use of the tangent plane is to obtain an approximation formula, which is a basic result in the study of functions of several variables.

The intuitive idea is that if we stay near the point (x_0, y_0, w_0) , the graph of the tangent plane provides a good approximation to the graph of the function $w = f(x, y)$.

Therefore, if the point (x, y) is close to (x_0, y_0) , then

$$f(x, y) \approx w_0 + \left(\frac{\partial w}{\partial x}\right)_0 (x - x_0) + \left(\frac{\partial w}{\partial y}\right)_0 (y - y_0).$$

Example 1

Solution

Using Equation (6), we calculate the two partial derivatives:

$$w_x = 3x^2y^4, \quad w_y = 4x^3y^3.$$

Evaluating the partial derivatives at $(1, 1)$ and using Equation (6), we obtain

$$\Delta w \approx 3\Delta x + 4\Delta y.$$

Thus, if

$$|\Delta x| \leq 0.01 \quad \text{and} \quad |\Delta y| \leq 0.01,$$

then

$$|\Delta w| \leq 3|\Delta x| + 4|\Delta y| \leq 0.07,$$

which is within the required bounds. Therefore, the answer is the square centered at $(1, 1)$ given by

$$|x - 1| \leq 0.01, \quad |y - 1| \leq 0.01.$$

Example 2

The sides a , b , and c of a rectangular box have lengths measured to be 1, 2, and 3, respectively. To which of these measurements is the volume V most sensitive?

Solution

$$V = abc,$$

and therefore, by the approximation formula (7),

$$\Delta w \approx \left(\frac{\partial w}{\partial x} \right)_0 \Delta x + \left(\frac{\partial w}{\partial y} \right)_0 \Delta y + \left(\frac{\partial w}{\partial z} \right)_0 \Delta z,$$

if

$$\Delta x, \Delta y, \Delta z \approx 0.$$

Unfortunately, for functions of three or more variables, we cannot use a geometric argument for the approximation formula (7). For this reason, it is best to recast the argument for (6) in a form that does not use tangent planes and geometry, and therefore can be generalized to several variables.

This is done at the end of this chapter. For now, let us simply assume the truth of (7) and its higher-dimensional analogue.

There are two typical examples of the approximation formula. Other examples appear in the exercises. In the rest of your study of partial differentiation, you will see how the approximation formula is used to derive important theorems and formulas.

Example 7

Give a reasonable square centered at $(1, 1)$ over which the value of

$$w = x^3 y^4$$

will not vary by more than ± 1 .

$$\Delta V \approx bc \Delta a + ac \Delta b + ab \Delta c$$

If

$$a = 6, \quad b = 3, \quad c = 3,$$

then

$$\Delta V \approx 9\Delta a + 18\Delta b + 18\Delta c.$$

Thus, it is the most sensitive to small changes in side a since Δa occurs with the largest coefficient. That is, if one side at a time changes by, say, 0.1, the change in a would produce the biggest change in V .

The result may seem paradoxical because the volume is most sensitive to the length of the shortest side, but it is actually intuitive when you think about the shape of the box.

Sensitivity Principle

The numerical value of

$$w = f(x, y, z, \dots)$$

calculated at some point

$$(x_0, y_0, z_0, \dots)$$

will be most sensitive to small changes in the variable whose corresponding partial derivative has the largest absolute value at that point.